US008838017B2

(12) **United States Patent**
Muthali et al.

(10) **Patent No.: US 8,838,017 B2**
(45) **Date of Patent: Sep. 16, 2014**

(54) **WIDEBAND JAMMER DETECTOR**

(75) Inventors: **Harish S. Muthali**, Austin, TX (US); **Shreyas Sen**, Atlanta, GA (US)

(73) Assignee: **QUALCOMM Incorporated**, San Diego, CA (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 284 days.

(21) Appl. No.: **12/582,090**

(22) Filed: **Oct. 20, 2009**

(65) **Prior Publication Data**

US 2010/0245151 A1 Sep. 30, 2010

Related U.S. Application Data

(60) Provisional application No. 61/165,090, filed on Mar. 31, 2009.

(51) **Int. Cl.**

H04K 3/00	(2006.01)
H03F 1/48	(2006.01)
H03F 1/08	(2006.01)
H04B 1/7097	(2011.01)
H04B 1/10	(2006.01)

(52) **U.S. Cl.**

CPC ***H03F 1/086*** (2013.01); *H04B 1/7097* (2013.01); ***H03F 1/483*** (2013.01); *H04B 1/1027* (2013.01)

USPC **455/1**; 455/550.1

(58) **Field of Classification Search**

USPC 455/127.2, 234.1, 134, 135, 214, 1

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

3,958,083	A	5/1976	Hara et al.
4,752,939	A	6/1988	Amoroso et al.
4,973,966	A	11/1990	Zeewy
5,121,202	A	6/1992	Tanoi
5,448,598	A	9/1995	Yousefi et al.
5,471,509	A	11/1995	Wood et al.

(Continued)

FOREIGN PATENT DOCUMENTS

CN	1266559	A	9/2000
CN	101366268	A	2/2009

(Continued)

OTHER PUBLICATIONS

Alberto Valdes-Garcia et al: "A Broadband CMOS Amplitude Detector for On-Chip RF Measurements" IEEE Transactions on Instrumentation and Measurement, IEEE Service Center, Piscataway, NJ, US, vol. 57, No. 7, Jul. 1, 2008, pp. 1470-1477, XP011205113 ISSN: 0018-9456 p. 1471, right-hand col.; figure 2.

(Continued)

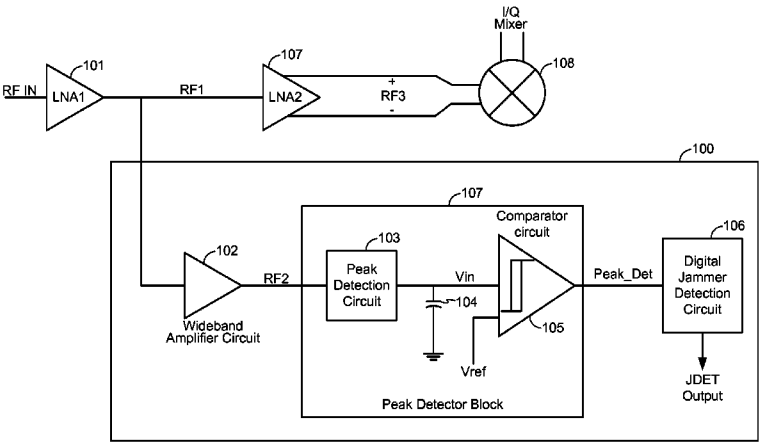
Primary Examiner — Tu X Nguyen

(74) *Attorney, Agent, or Firm* — Kevin T. Cheatham

(57) **ABSTRACT**

Techniques for detecting jammer signals in a received signal are described. In one aspect, high-speed current mirror resistive compensation circuits and output impedance boosting circuits are utilized to increase amplifier bandwidth in an improved wideband amplifier circuit. In another aspect, a dual transistor configuration including common source topology, averaging capacitors and a comparator circuit is utilized to improve the sensing of signal peaks in a peak detector block, which can be used together with the wideband amplifier circuit and a digital jammer detection circuit to detect jammer signals. The digital jammer detection circuit aids in the determination of the presence of jammer signals within the received signal, the determination of which may be variable due to programmability of the digital jammer detection circuit as described.

17 Claims, 7 Drawing Sheets



(56) **References Cited**

U.S. PATENT DOCUMENTS

6,429,696	B1	8/2002	Kao et al.	
6,980,786	B1	12/2005	Groe	
7,009,603	B2 *	3/2006	Page et al.	345/211
7,454,190	B2 *	11/2008	Schrodinger	455/341
2002/0101936	A1	8/2002	Wright et al.	
2004/0198284	A1	10/2004	Khorram	
2005/0130687	A1 *	6/2005	Filipovic et al.	455/522
2005/0215204	A1	9/2005	Wallace et al.	
2005/0221790	A1	10/2005	Persico et al.	
2005/0285781	A1 *	12/2005	Park et al.	342/357.02
2006/0255860	A1 *	11/2006	Moussavi	330/278
2007/0230610	A1	10/2007	Poberezhskiy	
2007/0232219	A1	10/2007	Xiong	
2007/0262817	A1 *	11/2007	Ciccarelli et al.	330/278
2008/0069013	A1	3/2008	Monier et al.	
2009/0066394	A1	3/2009	Kanda et al.	
2009/0128706	A1 *	5/2009	Nagasawa et al.	348/689
2010/0040178	A1	2/2010	Sutton et al.	
2010/0052968	A1	3/2010	Patel et al.	

FOREIGN PATENT DOCUMENTS

EP	0987825		3/2000
EP	1635240	A1	3/2006
JP	H08307924	A	11/1996
JP	H11239065	A	8/1999
JP	2001053630	A	2/2001
JP	2003069436	A	3/2003
JP	2004328639	A	11/2004
KR	20090005189	A	1/2009
TW	200807901	A	2/2008

OTHER PUBLICATIONS

International Search Report and Written Opinion—PCT/US2010/029517—International Search Authority, European Patent Office,Nov. 9, 2010.

Partial International Search Report—PCT/US2010/029517, International Search Authority—European Patent Office—Jul. 1, 2010.

Schlogl F et al: "120nm CMOS OPAMP with 690 MHz fT and 128 dB DC gain" Solid-State Circuits Conference, 2005. ESSCIRC 2005. Proceedings of the 31st European, IEEE, Piscataway, NJ, USA LNKDDOI: 10.1109/ESSCIR.2005.1541607, Sep. 12, 2005, pp. 251-254, XP010854950 ISBN: 978-0-7803-9205-2.

Tetsuro Itakura et al: "A Feedforward Technique With Frequency-Dependent Current Mirrors for a Low-Voltage Wideband Amplifier" IEEE Journal of Solid-State Circuits, IEEE Service Center, Piscataway, NJ, US LNKD- 001:10.1109/4.509872, vol. 31, No. 6, Jun. 1, 1996, pp. 847-849, XP000620189 ISSN: 0018-9200.

Voo, et al., High-speed current mirror resistive compensation technique, Electronics Letters, 1995, vol. 31, vol. 4, pp. 248-250.

* cited by examiner

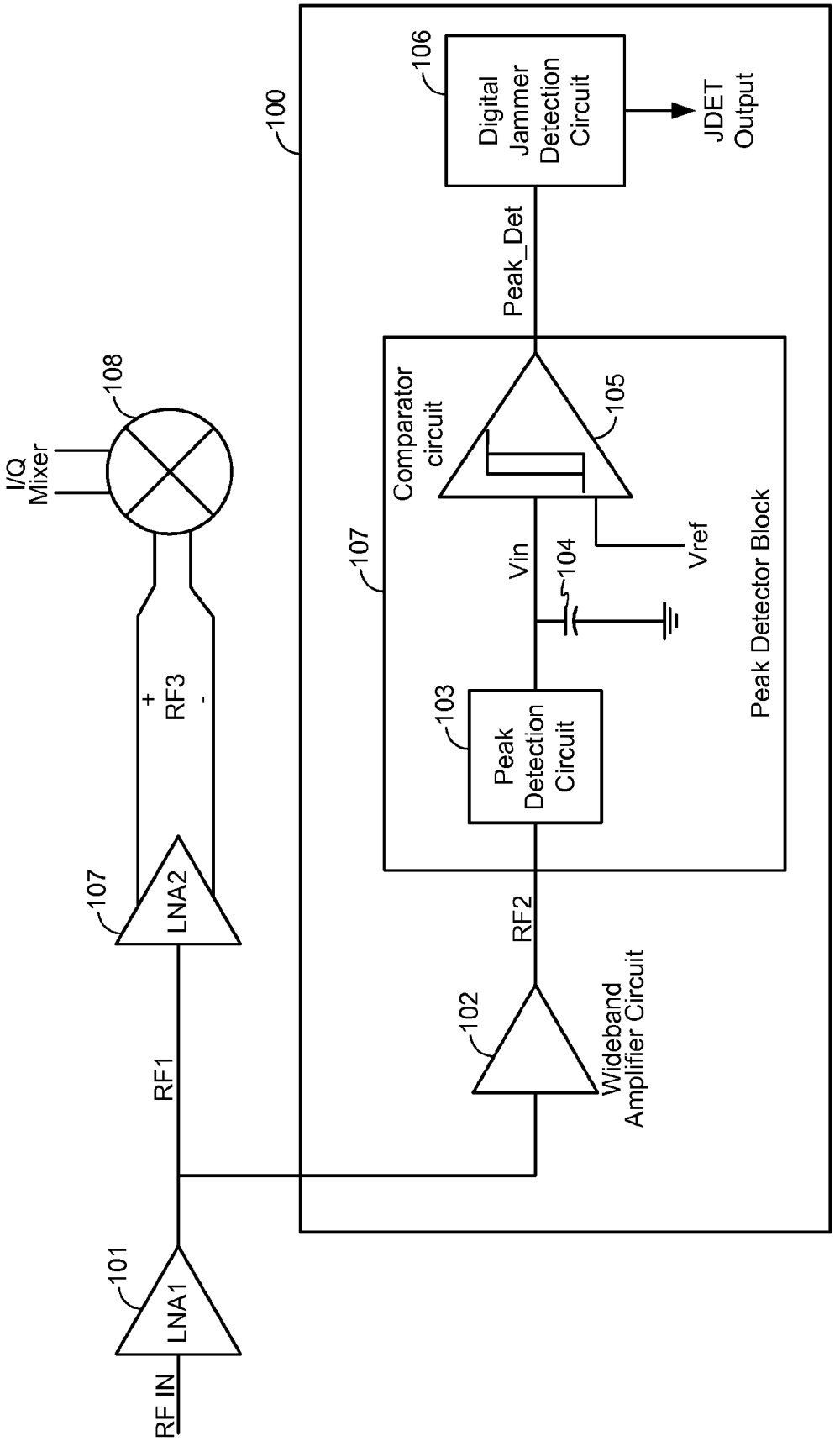

100
106
Digital Jammer Detection Circuit
JDET Output
Peak_Det
Comparator circuit
105
Vref
Vin
104
107
Peak Detector Block
103
Peak Detection Circuit
RF2
102
Wideband Amplifier Circuit
RF1
107
LNA2
+
RF3
-
108
I/Q Mixer
101
LNA1
RF IN

FIG. 1

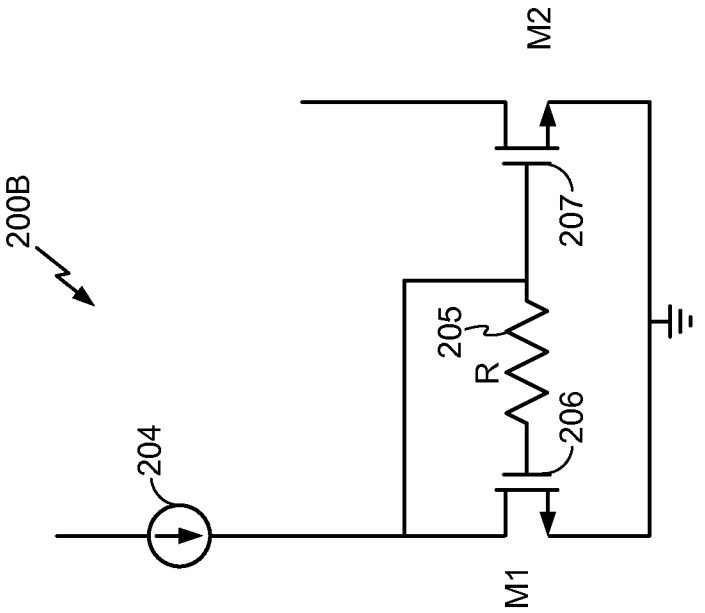


FIG. 2B

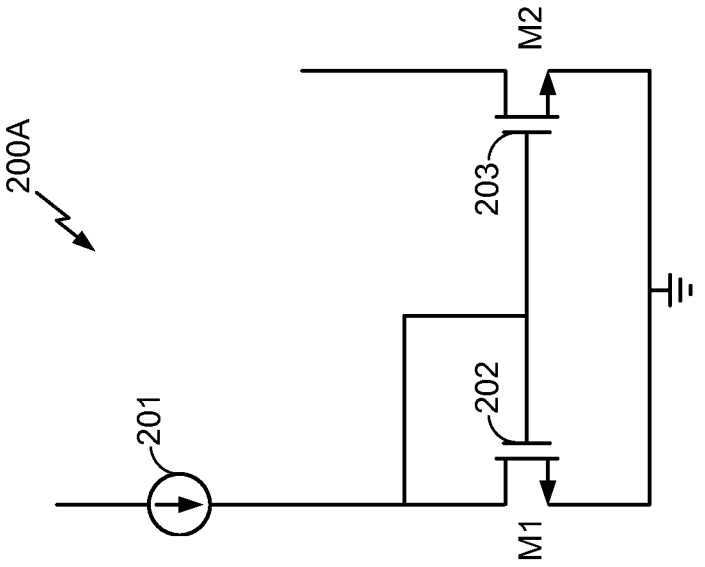


FIG. 2A

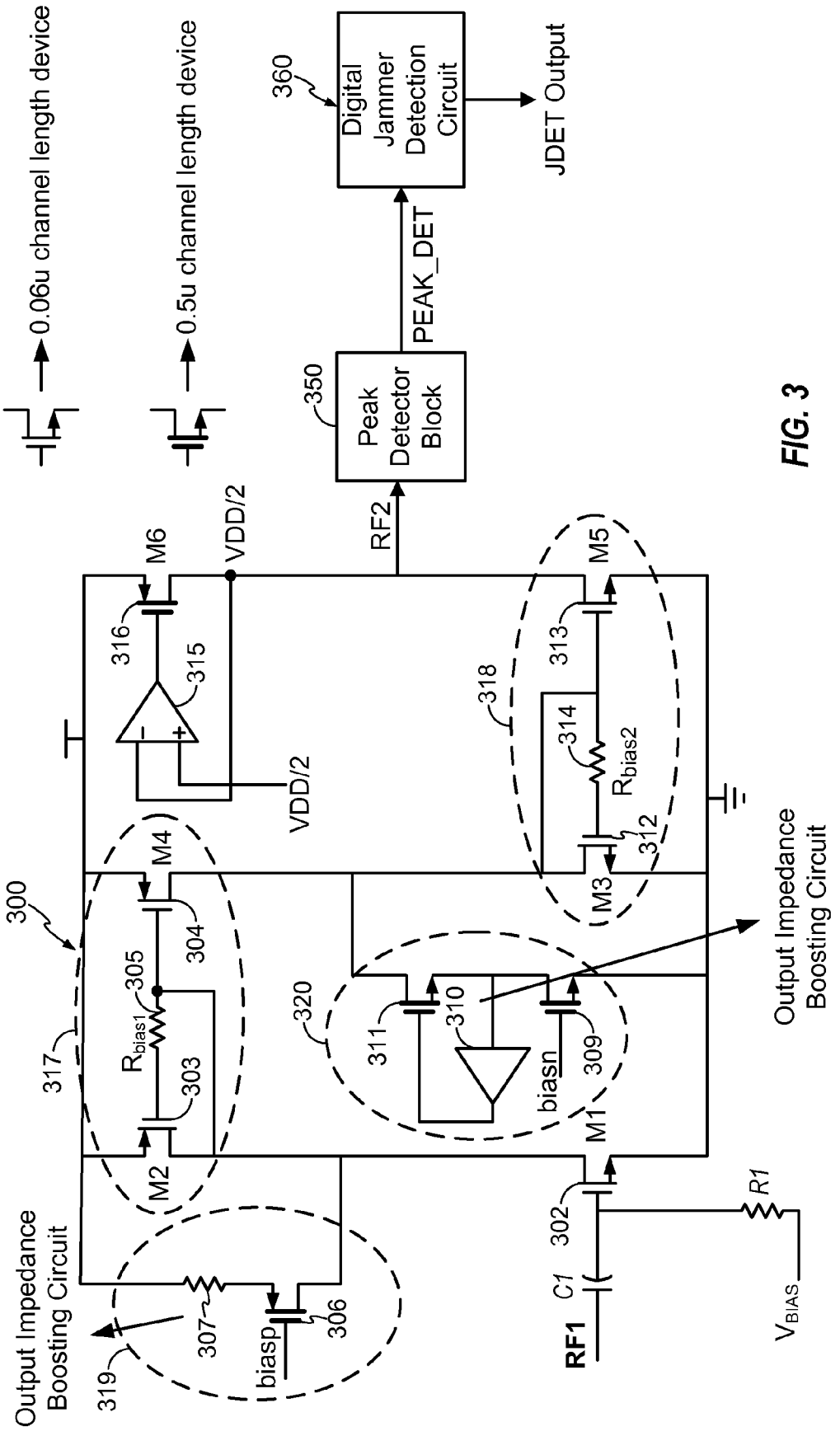

0.06u channel length device
0.5u channel length device
360
Digital Jammer Detection Circuit
JDET Output
PEAK_DET
350
Peak Detector Block
RF2
VDD/2
M6
316
315
VDD/2
M5
313
318
314
Rbias2
312
M3
M4
304
305
Rbias1
303
317
300
320
311
310
309
biasn
M1
Output Impedance Boosting Circuit
M2
307
biasp
306
319
Output Impedance Boosting Circuit
302
C1
RF1
R1
VBIAS

FIG. 3

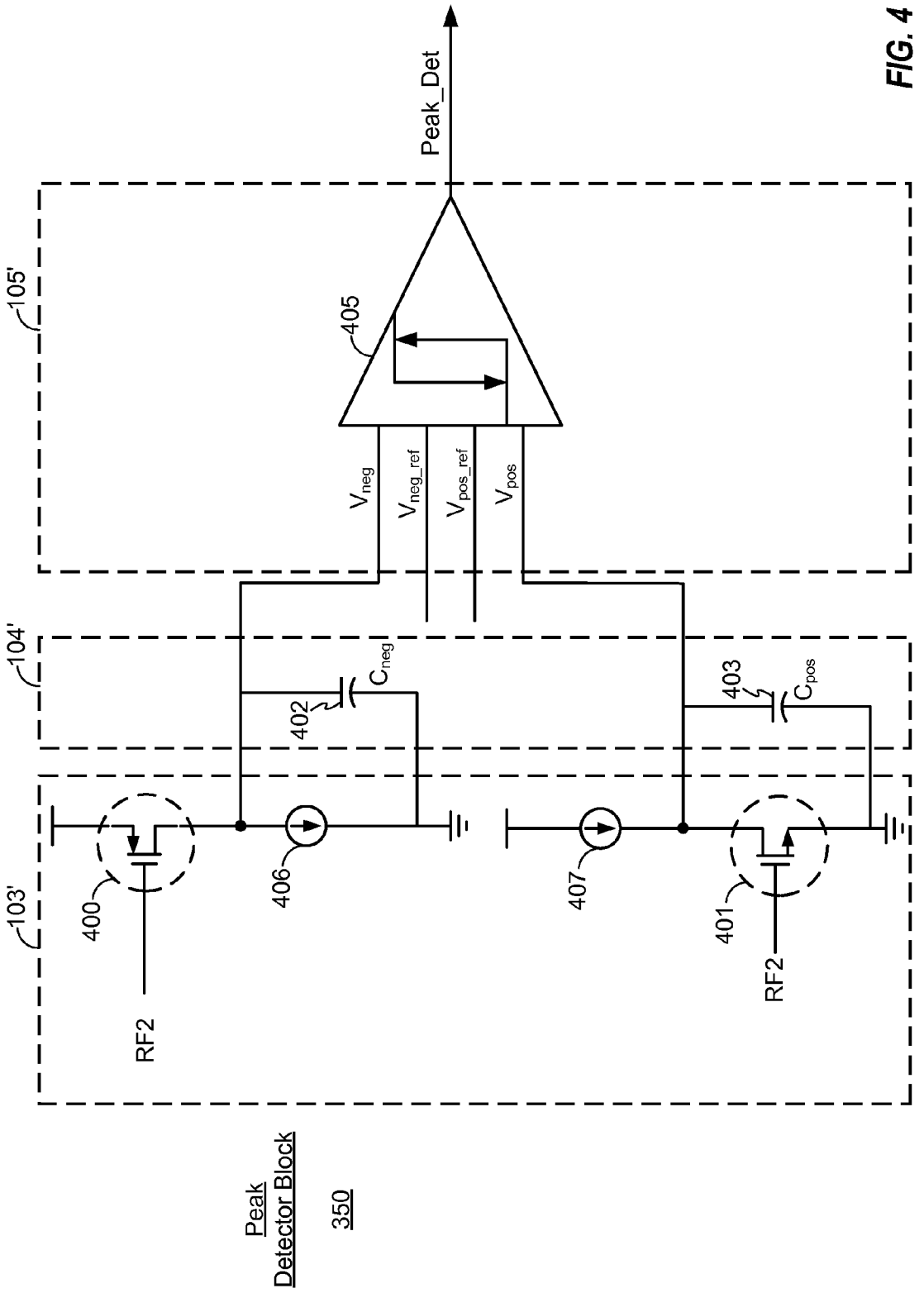

Peak_Det
105'
405
V_{neg}
V_{neg_ref}
V_{pos_ref}
V_{pos}
104'
402
C_{neg}
403
C_{pos}
103'
400
406
407
401
RF2
RF2
Peak Detector Block
350

FIG. 4

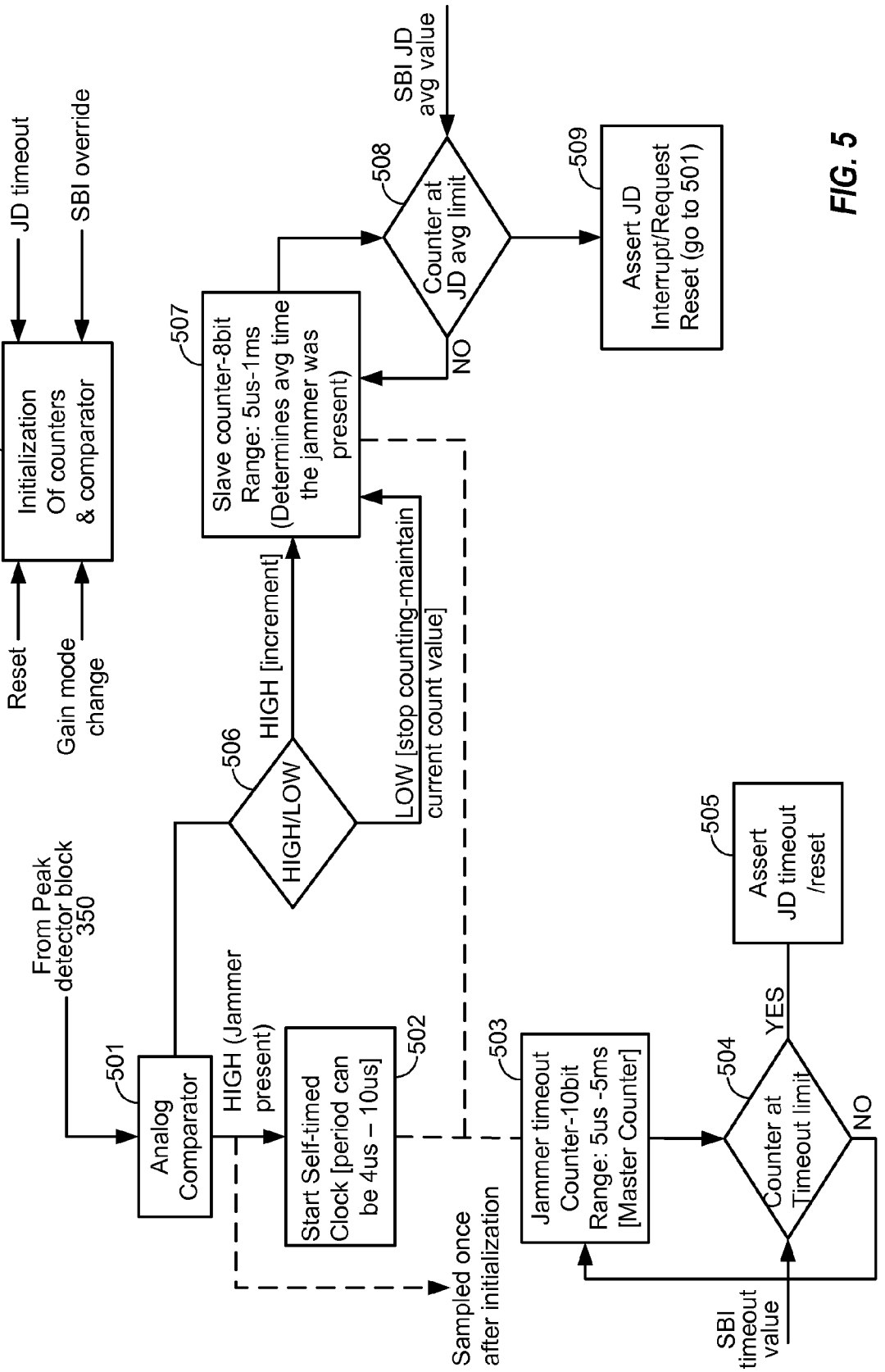

JD timeout
SBI override
500
Initialization
Of counters
& comparator
Reset
Gain mode
change
From Peak
detector block
350
501
Analog
Comparator
HIGH (Jammer
present)
Start Self-timed
Clock [period can
be 4us – 10us]
502
Sampled once
after initialization
503
Jammer timeout
Counter-10bit
Range: 5us -5ms
[Master Counter]
504
Counter at
Timeout limit
YES
NO
505
Assert
JD timeout
/reset
SBI
timeout
value
506
HIGH/LOW
HIGH [increment]
LOW [stop counting-maintain
current count value]
507
Slave counter-8bit
Range: 5us-1ms
(Determines avg time
the jammer was
present)
508
Counter at
JD avg limit
NO
SBI JD
avg value
509
Assert JD
Interrupt/Request
Reset (go to 501)

FIG. 5

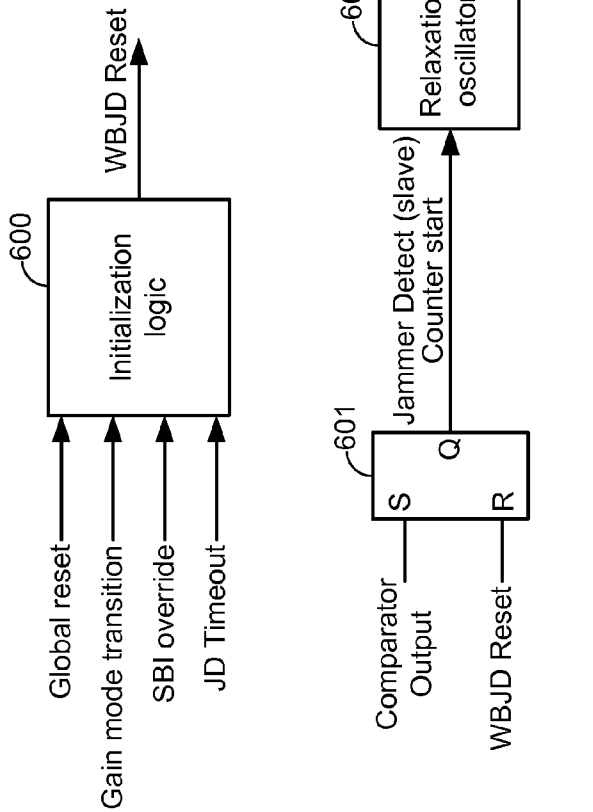

600
Initialization logic
WBJD Reset
Global reset
Gain mode transition
SBI override
JD Timeout
601
S
Q
R
Jammer Detect (slave) Counter start
Comparator Output
WBJD Reset
602
Relaxation oscillator
Jammer Detect CLK
603
8 bit counter
SBI compare
604
XOR
JD Interrupt

FIG. 6A

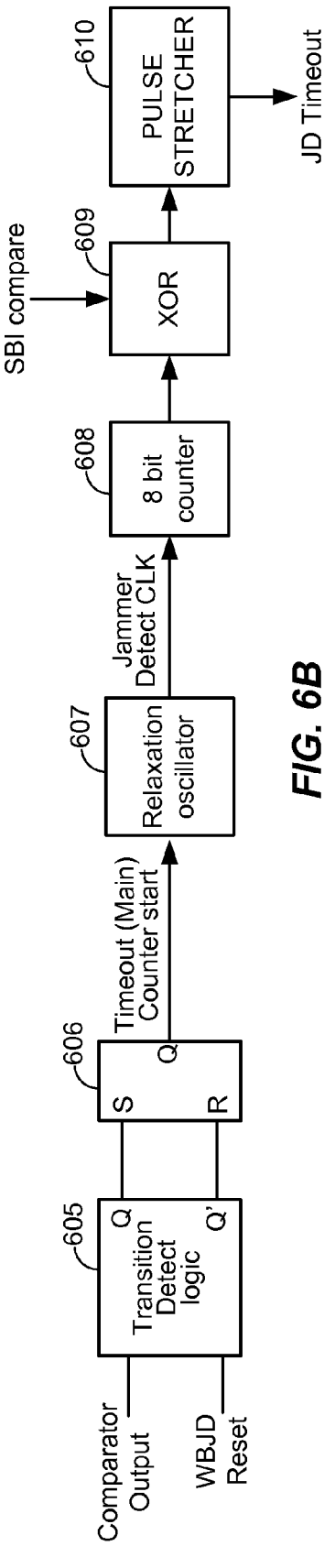

605
Transition Detect logic
Q
Q'
Comparator Output
WBJD Reset
606
S
R
Q
Timeout (Main) Counter start
607
Relaxation oscillator
Jammer Detect CLK
608
8 bit counter
SBI compare
609
XOR
610
PULSE STRETCHER
JD Timeout

FIG. 6B

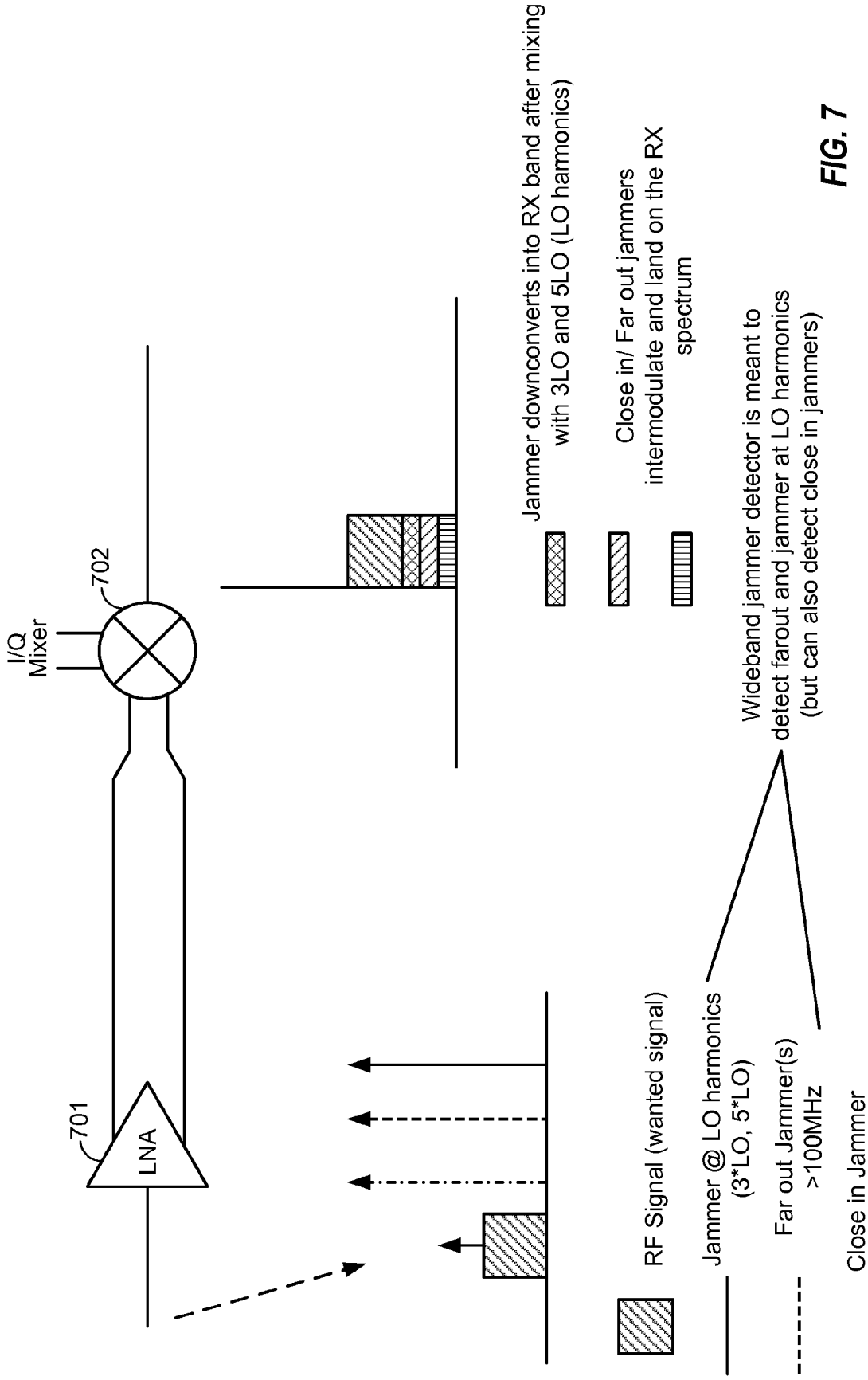

I/Q Mixer
702
701
LNA
Jammer downconverts into RX band after mixing with 3LO and 5LO (LO harmonics)
Close in/ Far out jammers intermodulate and land on the RX spectrum
Wideband jammer detector is meant to detect farout and jammer at LO harmonics (but can also detect close in jammers)
RF Signal (wanted signal)
Jammer @ LO harmonics (3*LO, 5*LO)
Far out Jammer(s) >100MHz
Close in Jammer (<40MHz)

FIG. 7

WIDEBAND JAMMER DETECTOR

CLAIM OF PRIORITY UNDER 35 U.S.C. §119

The present application for patent claims priority to Provisional Application No. 61/165,090 entitled, "Wideband Jammer Detector" filed Mar. 31, 2009, and assigned to the assignee hereof and hereby expressly incorporated by reference herein.

TECHNICAL FIELD

The present disclosure relates generally to electronics, and more specifically to a wideband jammer detector.

BACKGROUND

In communication devices, such as Code Division Multiple Access (CDMA) Global System for Mobile communications (GSM) and Wireless Local Area Network (WLAN) communication devices, the ability to detect jammer signals is necessary to improve the performance of the communication device. Communication devices include receiver circuits, which utilize correlation circuits to decipher a desired communication signal from all other received signals. Device performance is degraded when jammer signals are present during the correlation process.

Jammer signals can be introduced by internal or external sources. An internal jammer signal is a jammer signal which is introduced by the receiver. An example is a clock spur generated by a voltage controlled oscillator (VCO) within the receiver.

An external jammer signal is a jammer signal, which is introduced by a source external to the receiver. An example is a signal transmitted by a transmitter in another communication device that generates out-of-band emissions in the receive frequency band of the receiver.

A jammer signal impacts the sensitivity of a receiver in two ways. It can for example de-sense an analog-to-digital converter in the receiver thereby degrading its sensitivity. A jammer signal that appears at odd harmonics of a local oscillator signal in the receiver is down converted into the receive band to degrade signal sensitivity at baseband. Thus, errors may be introduced into the demodulated data packet when jammer signals are not detected by the receiver.

All jammer signals can thus degrade the performance of the receiver within the communication device, and ultimately the device's ability to process signals. Therefore, the ability to detect as many jammer signals as possible, even very low power jammer signals, helps to improve the performance of the receiver within the communication device.

In a typical CDMA device, for example, when a CDMA jammer signal is present, the receiver goes into protected mode where a decision is made by a jammer detector. The jammer detector detects close in jammer signals, such as jammer signals close to the receive (RX) band. A wideband jammer detector is capable of also detecting jammer signals hundreds of MHz away from the RX band. A jammer detector allows the receiver to operate in an un-protected or low power mode when no jammer signal is present and in protected or high power mode in the presence of jammer signals.

There is a need for a wideband jammer detector capable of detecting low power jammer signals over a wide bandwidth while consuming a minimal amount of power.

SUMMARY

This disclosure describes in general techniques for detecting jammer signals in a received signal.

In one aspect of the invention high-speed current mirror resistive compensation circuits and output impedance boosting circuits are utilized to increase amplifier bandwidth in an improved wideband amplifier circuit.

In another aspect of the invention, a dual transistor configuration including common source topology, averaging capacitors and a comparator circuit is utilized to improve the sensing of signal peaks in a peak detector block.

In another aspect of the invention, the peak detector block is used together with the wideband amplifier circuit and a digital jammer detection circuit to detect jammer signals.

In yet another aspect of the invention, the wideband amplifier circuit is used together with a peak detection circuit, a comparator circuit and a digital jammer detection circuit form a wideband jammer detector to detect jammer signals in a received signal of a communication device.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a block diagram of a receiver with a wideband jammer detector.

FIGS. **2**A and **2**B are circuit level diagrams of a standard current mirror resistive compensation circuit and a high-speed current mirror resistive compensation circuit, respectively.

FIG. **3** is a schematic diagram of a wideband amplifier circuit, a peak detector block and a digital jammer detection circuit in accordance with an exemplary embodiment.

FIG. **4** shows a schematic diagram of a peak detector block in accordance with a further exemplary embodiment which can be used to peak detect the output RF**2** from the wideband amplifier circuit in the standard receiver of FIG. **1** as well as a receiver incorporating the wideband amplifier circuit shown in FIG. **3**.

FIG. **5** shows an operational flow diagram of a digital jammer detection circuit **350** in accordance with an exemplary embodiment.

FIGS. **6**A and **6**B show two different digital logic configurations for implementing a digital jammer detection circuit in accordance with an exemplary embodiment.

FIG. **7** is a block diagram of a receiver path.

To facilitate understanding, identical reference numerals have been used where possible to designate identical elements that are common to the figures, except that suffixes may be added, when appropriate, to differentiate such elements. The images in the drawings are simplified for illustrative purposes and are not necessarily depicted to scale.

The appended drawings illustrate exemplary configurations of the disclosure and, as such, should not be considered as limiting the scope of the disclosure that may admit to other equally effective configurations. Correspondingly, it has been contemplated that features of some configurations may be beneficially incorporated in other configurations without further recitation.

DETAILED DESCRIPTION

The word "exemplary" is used herein to mean "serving as an example, instance, or illustration." Any embodiment described herein as "exemplary" is not necessarily to be construed as preferred or advantageous over other embodiments.

The detailed description set forth below in connection with the appended drawings is intended as a description of exemplary embodiments of the present invention and is not intended to represent the only embodiments in which the present invention can be practiced. The term "exemplary" used throughout this description means "serving as an example, instance, or illustration," and should not necessarily be construed as preferred or advantageous over other exemplary embodiments. The detailed description includes spe-

cific details for the purpose of providing a thorough understanding of the exemplary embodiments of the invention. It will be apparent to those skilled in the art that the exemplary embodiments of the invention may be practiced without these specific details. In some instances, well known structures and devices are shown in block diagram form in order to avoid obscuring the novelty of the exemplary embodiments presented herein. FIG. **1** is a block diagram of a standard receiver with a wideband jammer detector **100**. A radio frequency input signal RF IN is amplified by a first low noise amplifier (LNA) **101** to generate amplified signal RF**1**. Amplified signal RF **1** is coupled at an input to the wideband jammer detector **100** and also to an input of a second low noise amplifier **107**. A differential output signal RF**3** is generated by the second low noise amplifier **107** to in-phase and quadrature (I/Q) mixer **108**.

Wideband jammer detector **100** comprises wideband amplifier circuit **102**, peak detection circuit **103**, averaging capacitor **104**, comparator circuit **105**, and digital jammer detection circuit **106**. Wideband amplifier circuit **102** is a low power, wide bandwidth amplifier. Wideband amplifier circuit receives and amplifies RF**1** from LNA **101** to generate amplified signal RF**2**. RF**2** is connected to an input of peak detection circuit **103**. Peak detection circuit **103** generates an output voltage level proportional to a peak voltage of amplified signal RF**2**.

Capacitor **104** is coupled at one end to ground. The opposite end of capacitor **104** is coupled to the output of peak detection circuit **103** and to a sampling input Vin of comparator circuit **105**. Capacitor **104** serves to average the output of peak detection circuit **103**. Comparator circuit **105** also receives a reference input Vref. The output of comparator circuit **105** varies as a function of input signals Vin and Vref. When Vin is greater than or equal to Vref, the output PEAK_DET of comparator circuit **105** switches to a logic high state. Conversely, when Vin is less than Vref, the output PEAK_DET switches to a logic low state. Peak detection circuit **103**, capacitor **104** and comparator circuit **105** define a peak detector block **107**.

PEAK_DET is coupled to the input of digital jammer detection circuit **106**. Digital jammer detection circuit **106** samples the value of PEAK_DET over a programmed duration. The programmed duration may be controlled and varied during device operation. When digital jammer detection circuit **106** counts a programmed threshold number of logic level high samples within the programmed duration, digital jammer detection circuit **106** generates an interrupt output jammer detector signal JDET.

FIGS. **2**A and **2**B are circuit level diagrams of a standard current mirror circuit **200**A and a high-speed current mirror resistive compensation circuit **200**B, respectively.

A standard current mirror circuit **200**A as shown in FIG. **2**A comprises a current source **201** and two transistors **202** and **203**. Transistors **202** and **203** are NMOS devices.

Transistor **202** is configured as the reference transistor with the drain of transistor **202** coupled to the gates of both transistors **202** and **203**. A 3 db cutoff frequency for such standard current mirror configuration can be expressed as:

$$W_o = \frac{g_m}{2C_{gs}} \quad \text{Eq. (1)}$$

where, g_m is the transconductance of transistor **202**, C_{gs} is the gate to source capacitance of transistor **202**, and ω_o is the 3 dB cutoff frequency in radians.

By comparison, a high-speed current mirror resistive compensation circuit as shown in FIG. **2**B comprises a current source **204**, resistor **205**, and two NMOS transistors **206** and **207**. A first terminal of resistor **205** is coupled to the gate of transistor **206**. A second terminal of resistor **205** is coupled to the gate of transistor **207** and the drain of transistor **206**. Transistor **206** is configured as the reference transistor with the drain of transistor **206** coupled to the gate of transistor **207** and to the second terminal of resistor **205**. A 3 dB cutoff frequency for such a high-speed current mirror resistive compensation configuration can be expressed as:

$$H(s) = \frac{\omega_o}{Z}\left[\frac{s+Z}{s^2+2\zeta\omega_o s+\omega_o^2}\right] \quad \text{Eq. (2)}$$

$$\omega_o = \sqrt{\frac{g_{m1}}{RC_{gs1}C_{gs2}}} \quad \text{Eq. (3)}$$

$$Z = \frac{1}{RC_{gs1}} \quad \text{Eq. (4)}$$

$$\zeta = \frac{C_{gs1}+C_{gs2}}{2\sqrt{g_{m1}RC_{gs1}C_{gs2}}} \quad \text{Eq. (5)}$$

when:

$$R = \frac{1}{g_{m1}} \Rightarrow \omega_o \frac{g_{m1}}{C_{gs1}} \Rightarrow f_1(\text{theoretical}) \quad \text{Eq. (6)}$$

where, g_{m1} is the transconductance of transistor **202**, C_{gs1} is the gate to source capacitance of transistor **202**, R is resistor **205**, and ω_o is the 3 dB cutoff frequency in radians. As can be seen, the addition of resistor **205** increases the theoretical 3 dB cutoff frequency by a factor of 2 when resistor **205** has a resistive value equal to the reciprocal of the transconductance of transistor **206**. This results in a significant increase in bandwidth.

FIG. **3** is a schematic diagram of a wideband amplifier circuit **300**, a peak detector block **350**, and a digital jammer detection circuit **360** in accordance with an exemplary embodiment. Wideband amplifier circuit **300** comprises capacitor C**1**, resistor R**1**, PMOS input transistor **302**, first and second high-speed current mirror resistive compensation circuits **317**, **318**, operational amplifier **315**, NMOS transistor **316**, and first and second output impedance boosting circuits **319**, **320**. Peak detection block **350** may comprise peak detection circuit **103**, capacitor **104** and comparator circuit **105**, as in the standard receiver shown in FIG. **1**. Similarly, digital jammer detection circuit may comprise a circuit as in the digital jammer detection circuit **106** in the standard receiver shown in FIG. **1**.

Alternatively, peak detection block **350** and digital jammer detection circuit **360** may correspond to new digital logic circuits described below in connection with FIGS. **4** and **6**, respectively.

Referring back to FIG. **3**, high-speed current mirror resistive compensation circuit **317** includes resistor **305** and PMOS transistors **303** and **304**. High-speed current mirror resistive compensation circuit **318** includes resistor **314** and NMOS transistors **312** and **313**.

A first terminal of capacitor C**1** is coupled to the signal input RF**1**. A second terminal of capacitor C**1** is coupled to the gate of transistor **302**. The series coupling of the input signal RF**1** through capacitor C**1** provides AC-coupling of the input

signal RF1. The AC coupling capacitor C1 isolates the DC level of the previous stage and allows NMOS transistor **302** to be biased at a desired value Vbias. A first terminal of resistor R1 is coupled to a voltage source V_{bias}. A second terminal of resistor **301** is coupled to the gate of transistor **302**. Varying V_{bias} controls the bias voltage applied to the gate of transistor **302**. When additional gain is desired Vbias is increased to create an increased voltage between the gate and the source, V_{GS}, of transistor **302**.

The relationship between V_{GS} and the transistor drain current can be expressed as:

$$I_D = \frac{\mu_n C_{ox}}{2}\frac{W}{L}(V_{gs} - V_{th})^2(1 + \lambda V_{DS}) \quad \text{Eq. (7)}$$

where, μ_n is the charge-carrier effective mobility, W is the gate width, L is the gate length and C_{ox} is the gate oxide capacitance per unit area of transistor **302**. The relationship between drain current and transconductance can be expressed as:

$$g_m = \frac{2I_D}{V_{GS} - V_{th}} = \frac{2I_D}{V_{ov}} \quad \text{Eq. (8)}$$

The source of transistor **302** is coupled to ground. The drain of transistor **302** is coupled to an active load including high-speed current mirror resistive compensation circuit **317** and output impedance boosting circuit **319**. As described above, high-speed current mirror resistive compensation circuit **317** provides additional bandwidth by introducing a zero. The output impedance boosting circuit **319** increases the output resistance of transistor **302**. The increase in output resistance increases the gain provided by transistor **302**.

Output impedance boosting circuit **319** has two purposes. First, the output impedance boosting circuit diverts the DC current away from transistor **303**, thereby allowing transistor **303** to be a small low current device with high bandwidth. Transistors **304** and **303** have a device size ratio of 4:1. The DC current as well as the RF current is amplified going from transistors **303** to **304**. Second, output impedance boosting circuit **319** diverts the DC current away from the main path. However, output impedance boosting circuit **319** should not divert the RF signal from the main path. This is achieved by boosting the output impedance. The output impedance may be expressed as:

$$R_{OUT}=(1+g_mR)r_{OUT} \quad \text{Eq. (9)}$$

where, R is resistor **307** and r_{OUT} is resistor **306**.

The output of high-speed current mirror resistive compensation circuit **317** is coupled to high-speed current mirror resistive compensation circuit **318** and output impedance boosting circuit **320**. Similar to output impedance boosting circuit **319**, output impedance boosting circuit **320** has the same two purposes. The output impedance may be expressed as:

$$R_{OUT}=A(1+g_mr_{OUT})r_{OUT} \quad \text{Eq. (10)}$$

where, A is amplifier **310** and r_{OUT} is resistor **311**.

The output impedance boosting is achieved with a different technique. The output of high-speed current mirror resistive compensation circuit **318** is coupled to the drain of transistor **316** and the negative input of operational amplifier **315**. The positive input of operational amplifier **315** is coupled to a voltage source set to VDD/2. The operational amplifier **315** configuration drives the DC output of the wideband amplifier to VDD/2 by way of the virtual short effect between the inputs of an operational amplifier (OP AMP) **315**. OP AMP **315** controls the gate voltage of transistor **316** to create a current source to feed transistor **313**. OP AMP **315** ensures that the node RF2 is biased at VDD/2. This is achieved by driving the gate of the transistor M6 to whatever voltage in order that RF2 is set at VDD/2. RF2 feeds peak detector block **350**.

The gain from RF input at transistor **302** to output node RF2 can be expressed as:

$$Gain_{DC} = g_mM1\left(\frac{W4\times W5}{W2\times W3}\right)R_{OUT} \quad \text{Eq. (11)}$$

In general and as used herein, a gain may be (i) equal to one in linear unit, which is zero dB in logarithm unit, (ii) greater than one in linear unit, or (iii) less than one in linear unit. A gain of greater than one in linear unit corresponds to signal amplification and a positive gain (in dB). A gain of less than one in linear unit corresponds to signal attenuation and a negative gain (in dB). Attenuation is negative gain, so that an attenuation of x dB is equivalent to a gain of −x dB.

FIG. **4** shows a schematic diagram of a peak detector block **350** in accordance with a further exemplary embodiment which can be used to peak detect the output RF2 from wideband amplifier circuit **102** in the standard receiver of FIG. **1** as well as a receiver incorporating the wideband amplifier circuit **300** shown in FIG. **3**.

Peak detector block **350** includes a peak detection circuit **103'**, a capacitor coupling circuit **104'** and a comparator circuit **105'**. Peak detection circuit **103'** includes an upper negative-peak detector portion and a lower positive-peak detector portion. The RF2 amplified output from a preceding stage is coupled to respective peak detector input transistors **400** and **401** in each of upper negative-peak and lower positive-peak detector portions. Transistor **400** is a PMOS transistor. The source of transistor **400** is coupled to VDD. PMOS transistor **400** is biased in weak inversion region to allow it to follow the negative peak of an amplified jammer signal. The drain of transistor **400** is coupled to current source **406**. PMOS transistor **400** is configured in a common source configuration to allow detection of negative peak signals. A first terminal of capacitor **402** is coupled to the drain of transistor **400** and the negative sampling input V_{neg} of comparator **405**. A second terminal of capacitor **402** is coupled to ground. Capacitor **402** averages the output of transistor **400** to create signal V_{neg}. The negative threshold input V_{neg_ref} is set to a selected threshold voltage level. In some designs the selected threshold voltage level V_{pos_ref} may be programmable.

Transistor **401** is a NMOS transistor biased to operate in the weak inversion region so the transistor will follow the positive peak of the amplified jammer signal. The source of transistor **401** is coupled to ground. The drain of transistor **401** is coupled to current source **407**. Thus, NMOS transistor **401** is configured in a common source configuration to allow detection of positive peak signals. A first terminal of capacitor **403** is coupled to the drain of transistor **401** and the positive sampling input of comparator **405**. A second terminal of capacitor **403** is coupled to ground. Capacitor **403** averages the output of transistor **401** to create signal V_{pos}. The positive threshold input value V_{pos_ref} is set to a selected threshold voltage level. In some designs, the selected threshold voltage level V_{pos_ref} may be programmable.

Transistors **400** and **401** are biased to operate in weak inversion mode, otherwise known as "Cut-off" or "Sub-threshold" mode. Weak inversion occurs when the gate to

source voltage is less than the threshold voltage of the transistor. Ideally, current should not flow through a transistor in weak inversion mode. However, due to the Boltzman distribution of electron energies some more energetic electrons at the source can enter the channel and flow to the drain of the transistor. This results in a sub-threshold current that is exponentially related to the gate to source voltage applied to the transistor. Operation in weak inversion mode allows transistors **400** and **401** to create an output current large enough to properly drive comparator **405**. The relationship between gate to source voltage and sub threshold current may be expressed as:

$$I_D = I_{DO} e^{\frac{V_{GS}-V_{th}}{nV_T}} \quad \text{Eq. (12)}$$

where, I_{D0}=current at $V_{GS}=V_{th}$ and the slope factor n is given by

$$n=1+C_D/C_{OX} \quad \text{Eq. (13)}$$

where, C_D=capacitance of the depletion layer and C_{OX}=capacitance of the oxide layer.

Comparator **405** compares the amplitude of input signal V_{neg} with the amplitude of threshold signal V_{neg_ref}. Comparator **405** also compares the amplitude of input signal V_{pos} with the amplitude of threshold signal V_{pos_ref}. When either input signal has larger amplitude than the amplitude of the corresponding threshold signal, the comparator **405** output signal PEAK_DET is set to a logic high state. When both input signals have lower amplitude than the amplitude of the corresponding threshold signal the comparator **405** output signal PEAK_DET is set to a logic low state.

As described in FIG. **1**, a conventional digital jammer detection circuit **106** samples the output of comparator **105** over a programmed duration. The programmed duration may be controlled and varied during device operation. When the digital jammer detection circuit **106** counts more than a programmed threshold number of logic level high samples within the programmed duration, digital detection circuit **106** generates an interrupt output signal JDET.

FIG. **5** shows an operational flow diagram of a digital jammer detection circuit **350** in accordance with an exemplary embodiment.

In step **500**, the comparator circuit **150** and all counters are initialized. Step **500** may be initiated by different events, such as: powering on the device, jammer detection circuit timeout, single wire bus interface (SBI) override, global reset, or gain mode transition. In step **500**, a digital jammer detection circuit clock is reset. Once the digital jammer detection circuit clock is reset in step **500**, two independent processes begin to operate in parallel. The first independent process includes steps **503**, **504**, and **505**. The second independent process includes steps **506**, **507**, **508** and **509**.

The first independent process includes steps **503**, **504**, and **505**. In step **503**, the digital jammer detection circuit timeout period is provided. In step **504**, the digital jammer detection circuit determines whether the jammer detection circuit timeout period provided in step **503** has elapsed. If the digital jammer detection circuit timeout period has not elapsed step **504** is repeated. In step **505**, a reset request is sent if the jammer timeout period has elapsed.

The second independent process includes steps **506**, **507**, **508** and **509**. In step **506** the digital jammer detection circuit samples the output of comparator **105** and determines if the sampled output is a logic high signal. If the sampled output is not a logic high signal the counting is stopped and current count value is maintained. In step **507**, the slave counter determines the average time the jammer was present. In step **508**, the digital jammer detection circuit determines if the peak counter value is greater than the programmed peak threshold value. In step **509**, a jammer interrupt signal is sent and a reset request generated sending the process back to step **501**.

The digital jammer detection circuit determines if an external reset request has been made. An external interrupt may be initiated by different events, such as: single wire bus interface (SBI) overwrite, global device reset, or gain mode transition.

FIGS. **6**A and **6**B show two different digital logic configurations for implementing digital jammer detection circuit **360** in accordance with an exemplary embodiment. Block **600** is the initialization logic, which generates a digital jammer detection circuit reset signal when any of the following inputs are triggered: global reset, gain mode transition, SBI override, or digital jammer detection circuit timeout.

In FIG. **6**A, block **601** is an SR latch circuit. An SR latch circuit is an arrangement of logic gates that maintains a stable output after the inputs have been turned off. A SR latch circuit has a set input (S) and a reset input (R). When the set input is logic high state the output is set to a logic high state. When the reset input is a logic high state the output is set to a logic low state. The output of block **601** is coupled to the input of block **602**.

Block **602** is a relaxation oscillation circuit. A relaxation oscillation circuit is an oscillator circuit that utilizes a capacitor, which is charged gradually and then discharged rapidly. A relation circuit may be implemented with a resistor or current source, a capacitor, and a threshold device such as an injunction transistor or Gunn diode. When the output of block **601** is ON the relaxation oscillator **602** creates an output signal, which oscillates at a predetermined frequency.

The oscillating output signal is the digital jammer detection circuit clock. The relaxation oscillator output is coupled to the input of the 8-bit counter **603**. The 8-bit counter **603** counts the oscillations sampled. 8-bit counter **603** outputs the number of oscillations, which have been observed. A first input to XOR circuit **604** is coupled to the output of the 8-bit counter **603**. A second input of XOR circuit **604** is coupled to a SBI compare signal. XOR circuit **604** determines a threshold compare value from the SBI compare signal. XOR circuit **604** compares the number provided by the 8-bit counter **603** with the SBI compare value. If the number provided by the 8-bit counter **603** is greater than the SBI compare value the XOR circuit **604** generates a jammer detected interrupt signal.

In FIG. **6**B, block **605** is a transition detect logic. Transition detect logic **605** detects any change or transition of the logic level of the input signals and generates a constantly high logic level on its corresponding output. The Q output of transition detect logic **605** is coupled to the S input of SR latch **606**. The Q' output (inverse of Q output) of transition detect logic **605** is coupled to the R input of SR latch **606**. The transition detector purpose is to detect the comparator output going high when a jammer signal is first present after the circuit is initialized. This starts the "scanning" mode when the circuit starts scanning for jammer signals. The transition detector triggers the timeout counter, which determines the time for which the circuit needs to scan for a jammer signal. If the jammer count reaches the desired threshold (determined by slave counter) before the timeout counter triggers then the JDET signal goes HIGH indicating jammer signal detection.

Block **606** is an SR latch circuit. An SR latch circuit is an arrangement of logic gates that maintains a stable output after the inputs have been turned off. An SR latch circuit has a set

input (S) and a reset input (R). When the set input is logic high state the output is set to a logic high state. When the reset input is a logic high state the output is set to a logic low state. The output of block **606** is coupled to the input of block **607**.

Block **607** is a relaxation oscillator. A relaxation oscillator is an oscillator circuit that utilizes a capacitor, which is charged gradually and then discharged rapidly. A relaxation oscillator may be implemented with a resistor or current source, a capacitor, and a threshold device such as a unijunction transistor or Gunn diode. When the output of block **606** is a logic high signal the relaxation oscillator **607** creates an output signal which oscillates at a predetermined frequency. The oscillating output signal is the digital jammer detection circuit clock. The relaxation oscillator output is coupled to the input of the 8-bit counter **608**. The 8-bit counter **608** counts the oscillations sampled. 8-bit counter **608** outputs the number of oscillations, which have been observed.

A first input to XOR circuit **609** is coupled to the output of 8-bit counter circuit **608**. A second input of XOR circuit **609** is coupled to a SBI compare signal. XOR circuit **609** determines a threshold compare value from the SBI compare signal. XOR circuit **609** compares the number provided by the 8-bit counter **608** with the SBI compare value. If the number provided by the 8-bit counter **608** is greater than the SBI compare value the XOR circuit **609** generates a jammer detected interrupt signal. The output of XOR circuit **609** is coupled to the input of pulse stretcher circuit **610**. Pulse stretcher **610** generates a extended reset signal, which is meant to reset the internal states/counters when a jammer timeout occurs

FIG. **7** shows a block diagram of a receiver path. The differential output terminals of LNA **701** are coupled to I/Q mixer **702**. The input signal coupled to the input of LNA **701** contains the desired RF signal, local oscillator (LO) signals, close in jammer signals and far out jammer signals. In accordance with the exemplary embodiments described herein, wideband jammer detector **100** detects the presence of LO signals, close in jammer signals and far out jammer signals.

Those of skill in the art would understand that information and signals may be represented using any of a variety of different technologies and techniques. For example, data, instructions, commands, information, signals, bits, symbols, and chips that may be referenced throughout the above description may be represented by voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, or any combination thereof.

Those of skill would further appreciate that the various illustrative logical blocks, modules, circuits, and algorithm steps described in connection with the embodiments disclosed herein may be implemented as electronic hardware, computer software, or combinations of both. To clearly illustrate this interchangeability of hardware and software, various illustrative components, blocks, modules, circuits, and steps have been described above generally in terms of their functionality. Whether such functionality is implemented as hardware or software depends upon the particular application and design constraints imposed on the overall system. Skilled artisans may implement the described functionality in varying ways for each particular application, but such implementation decisions should not be interpreted as causing a departure from the scope of the exemplary embodiments of the invention.

The various illustrative logical blocks, modules, and circuits described in connection with the embodiments disclosed herein may be implemented or performed with a general purpose processor, a Digital Signal Processor (DSP), an Application Specific Integrated Circuit (ASIC), a Field Programmable Gate Array (FPGA) or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general purpose processor may be a microprocessor, but in the alternative, the processor may be any conventional processor, controller, microcontroller, or state machine. A processor may also be implemented as a combination of computing devices, e.g., a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration.

The steps of a method or algorithm described in connection with the embodiments disclosed herein may be embodied directly in hardware, in a software module executed by a processor, or in a combination of the two. A software module may reside in Random Access Memory (RAM), flash memory, Read Only Memory (ROM), Electrically Programmable ROM (EPROM), Electrically Erasable Programmable ROM (EEPROM), registers, hard disk, a removable disk, a CD-ROM, or any other form of storage medium known in the art. An exemplary storage medium is coupled to the processor such that the processor can read information from, and write information to, the storage medium. In the alternative, the storage medium may be integral to the processor. The processor and the storage medium may reside in an ASIC. The ASIC may reside in a user terminal. In the alternative, the processor and the storage medium may reside as discrete components in a user terminal.

In one or more exemplary embodiments, the functions described may be implemented in hardware, software, firmware, or any combination thereof. If implemented in software, the functions may be stored on or transmitted over as one or more instructions or code on a computer-readable medium. Computer-readable media includes both computer storage media and communication media including any medium that facilitates transfer of a computer program from one place to another. A storage media may be any available media that can be accessed by a computer. By way of example, and not limitation, such computer-readable media can comprise RAM, ROM, EEPROM, CD-ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other medium that can be used to carry or store desired program code in the form of instructions or data structures and that can be accessed by a computer. Also, any connection is properly termed a computer-readable medium. For example, if the software is transmitted from a website, server, or other remote source using a coaxial cable, fiber optic cable, twisted pair, digital subscriber line (DSL), or wireless technologies such as infrared, radio, and microwave, then the coaxial cable, fiber optic cable, twisted pair, DSL, or wireless technologies such as infrared, radio, and microwave are included in the definition of medium. Disk and disc, as used herein, includes compact disc (CD), laser disc, optical disc, digital versatile disc (DVD), floppy disk and blu-ray disc where disks usually reproduce data magnetically, while discs reproduce data optically with lasers. Combinations of the above should also be included within the scope of computer-readable media.

The previous description of the disclosed exemplary embodiments is provided to enable any person skilled in the art to make or use the present invention. Various modifications to these exemplary embodiments will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other embodiments without departing from the spirit or scope of the invention. Thus, the present invention is not intended to be limited to the embodi-

ments shown herein but is to be accorded the widest scope consistent with the principles and novel features disclosed herein.

What is claimed is:

1. A peak detector block comprising: a first transistor, a second transistor, a first capacitor, a second capacitor and a comparator circuit, wherein:

a gate terminal of the first transistor is coupled to a signal input;

a source terminal of the first transistor is coupled to a first power source;

a drain terminal of the first transistor is coupled to a first terminal of the first capacitor and a first input terminal of the comparator circuit;

a second terminal of the first capacitor is coupled to a ground source;

a gate terminal of the second transistor is coupled to the signal input;

a source terminal of the second transistor is coupled to the ground source;

a drain terminal of the second transistor is coupled to a first terminal of the second capacitor and a second input terminal of the comparator circuit, wherein the comparator circuit is configured to compare a first signal provided to the first input terminal of the comparator to a first threshold signal and to compare a second signal provided to the second input terminal of the comparator to a second threshold signal; and

a second terminal of the second capacitor is coupled to the ground source.

2. The peak detector block of claim **1**, where at least one of the first transistor and the second transistor is biased to operate in weak inversion mode.

3. The peak detector block of claim **1**, where the first transistor is a PMOS transistor and the second transistor is a NMOS transistor.

4. The peak detector block of claim **1**, where the first and second transistors are coupled in a common source configuration.

5. The peak detector block of claim **1**, where each of the first and second threshold signals is programmable.

6. The peak detector block of claim **1**, wherein the drain terminal of the first transistor is coupled to a current source.

7. The peak detector block of claim **6**, wherein the drain terminal of the second transistor is coupled to a second current source.

8. The peak detector block of claim **1**, wherein the comparator circuit is configured to generate an output signal having a first value in response to the first signal having a larger amplitude than an amplitude of the first threshold signal, and wherein the comparator circuit is configured to generate the output signal having the first value in response to the second signal provided to the second input terminal of the comparator having a larger amplitude than an amplitude of the second threshold signal.

9. A wideband jammer detector including:

a wideband amplifier circuit;

a peak detection circuit;

a comparator circuit; and

a digital jammer detection circuit, wherein:

a signal input is coupled to an input terminal of the wideband amplifier circuit;

an output terminal of the wideband amplifier circuit is coupled to an input terminal of the peak detection circuit;

an output terminal of the peak detection circuit is coupled to a first input terminal of the comparator circuit;

a threshold signal input is coupled to a second input terminal of the comparator circuit;

an output of the comparator circuit is coupled to an input terminal of the digital jammer detection circuit; and

the wideband amplifier circuit includes a first active load and a second active load, wherein the first active load is coupled to a drain terminal of an input transistor, wherein the first active load is configured as a high-speed current mirror resistive compensation circuit, wherein the second active load is coupled to the drain terminal of the input transistor, and wherein the second active load is configured as an output impedance boosting circuit.

10. The wideband jammer detector of claim **9**, wherein a programmable threshold signal is provided to the second input terminal of the comparator circuit via the threshold signal input.

11. The wideband jammer detector of claim **9**, wherein the wideband amplifier circuit further includes a third active load coupled to a output of the first active load configured as a second high-speed current minor resistive compensation circuit, and a fourth active load coupled to the output of the first active load and configured as a second output impedance boosting circuit.

12. The wideband jammer detector of claim **9**, wherein the high-speed current mirror resistive compensation circuit includes:

a first transistor;

a second transistor; and

a first resistor, wherein:

a drain terminal of the first transistor is coupled to the drain terminal of the input transistor;

a source terminal of the first transistor is coupled to a gate terminal of the second transistor;

a first terminal of the first resistor is coupled to the gate terminal of the first transistor;

a second terminal of the first resistor is coupled to a gate terminal of the second transistor;

a source terminal of the first transistor is coupled to a power source; and

a source terminal of the second transistor is coupled to the power source.

13. The wideband jammer detector of claim **9**, wherein the peak detection circuit includes a first transistor, a second transistor, a first capacitor, a second capacitor, wherein the comparator circuit includes a comparator, and wherein:

a gate terminal of the first transistor is coupled to a signal input;

a source terminal of the first transistor is coupled to a first power source;

a drain terminal of the first transistor is coupled to a first terminal of the first capacitor and to a first input terminal of the comparator;

a second terminal of the first capacitor is coupled to a ground source;

a gate terminal of the second transistor is coupled to the signal input;

a source terminal of the second transistor is coupled to the ground source;

a drain terminal of the second transistor is coupled to a first terminal of the second capacitor and to a second input terminal of the comparator; and

a second terminal of the second capacitor is coupled to the ground source.

14. The wideband jammer detector of claim 9, wherein the digital jammer detection circuit comprises:

means for amplifying a received signal;

means for measuring a peak of the received signal;

means for comparing the peak of the received signal with a threshold signal;

means for counting the number of peaks which are greater in magnitude than the threshold signal during a desired duration; and

means for generating an interrupt signal if a counted number of peaks is greater than a threshold value.

15. An apparatus comprising:

means for amplifying a signal;

means for detecting a peak of the amplified signal;

means for comparing the peak of the amplified signal to a threshold signal;

means for detecting a digital jammer signal, wherein:

a signal input is coupled to an input terminal of the means for amplifying;

an output terminal of the means for amplifying the signal is coupled to an input terminal of the means for detecting the peak of the amplified signal;

an output terminal of the means for detecting the peak of the amplified signal is coupled to a first input terminal of the means for comparing;

a threshold signal input is coupled to a second input terminal of the means for comparing;

an output terminal of the means for comparing is coupled to an input terminal of the means for detecting a digital jammer signal; and

the means for amplifying includes a first active load and a second active load, wherein the first active load is coupled to a drain terminal of an input transistor, wherein the first active load is configured as a high-speed current mirror resistive compensation circuit, wherein the second active load is coupled to the drain terminal of the input transistor, and wherein the second active load is configured as an output impedance boosting circuit.

16. The apparatus of claim 15, wherein the threshold signal is programmable.

17. The apparatus of claim 15, wherein the means for amplifying further includes a third active load and a fourth active load wherein the third active load is coupled to an output of the first active load, wherein the third active load is configured as a second high-speed current minor resistive compensation circuit, wherein the fourth active load is coupled to the output of the first active load, and wherein the fourth active load is configured as a second output impedance boosting circuit.

* * * * *